Modeling, Risk & Results Report

Supply Chain Optimization Project

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1. Objective

The objective of this phase is to evaluate delivery performance, build demand forecasting models, and apply inventory optimization techniques to improve operational efficiency and reduce supply chain cost. This stage converts exploratory data insights into quantitative models and actionable decisions.

The modeling stage focuses on:

- Delivery performance and logistics risk
- Demand forecasting using time-series methods
- Inventory optimization using EOQ
- Safety stock and reorder point design
- ABC classification and cost impact
- Risk and assumption evaluation

2. Delivery Performance Analysis

Delivery performance was evaluated using purchase, estimated delivery, and actual delivery timestamps. Two derived metrics were created:

- **Delivery Time** = Delivered Date − Purchase Date
- **Delivery Delay** = Delivered Date − Estimated Date

These metrics help measure logistics reliability and customer service level.

2.1 Delivery KPIs

(Replace XXX with your computed numbers)

- Average Delivery Time: **12.00 days**
- Late Delivery Percentage: **6.59 %**
- On-Time Delivery Percentage: **1.31%**
- Early Delivery Percentage: **92.02 %**

2.2 Delay Distribution Insight

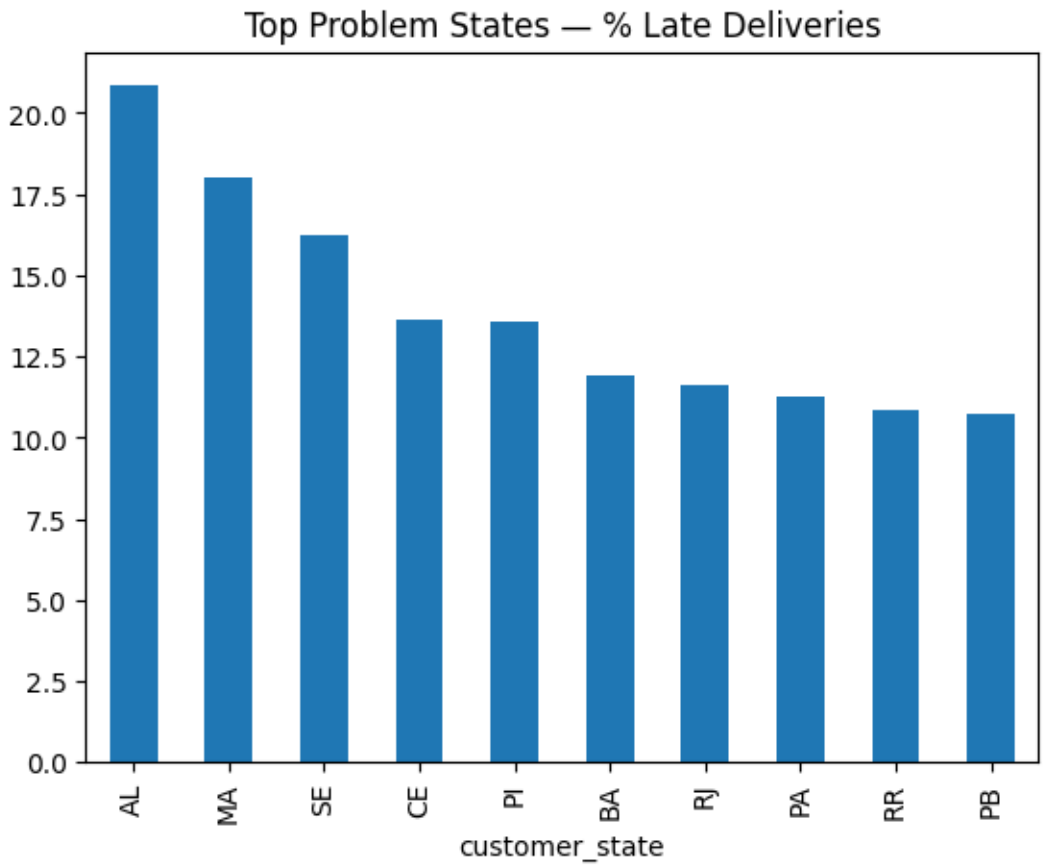
The delay distribution shows that most deliveries occur near the estimated date, but a noticeable tail of delayed deliveries exists. This indicates operational variability and route-based inefficiencies.

3. Regional Delivery Risk

Delivery performance varies significantly across regions.

Key observations:

- Certain customer states show higher late percentages
- Cross-state shipments have longer delivery times
- Some seller–customer state pairs act as bottleneck routes



3.1 Bottleneck Route Analysis

Route-level grouping of seller state to customer state reveals specific high-risk corridors where delays are concentrated. These routes should be prioritized for logistics improvement and carrier optimization.

4. Demand Forecasting Models

Daily demand time series were constructed for top product categories. Forecasting was performed using baseline time-series models.

Models used:

- Moving Average
- Exponential Smoothing

4.1 Moving Average Model

The moving average model smooths short-term fluctuations by averaging recent demand values. It reduces noise but reacts slowly to sudden level changes.

Strength:

- Simple
- Stable smoothing

Limitation:

- Lag in trend response

4.2 Exponential Smoothing Model

Exponential smoothing assigns higher weight to recent demand and adapts faster to demand level shifts.

Strength:

- Faster adaptation
- Better short-term accuracy

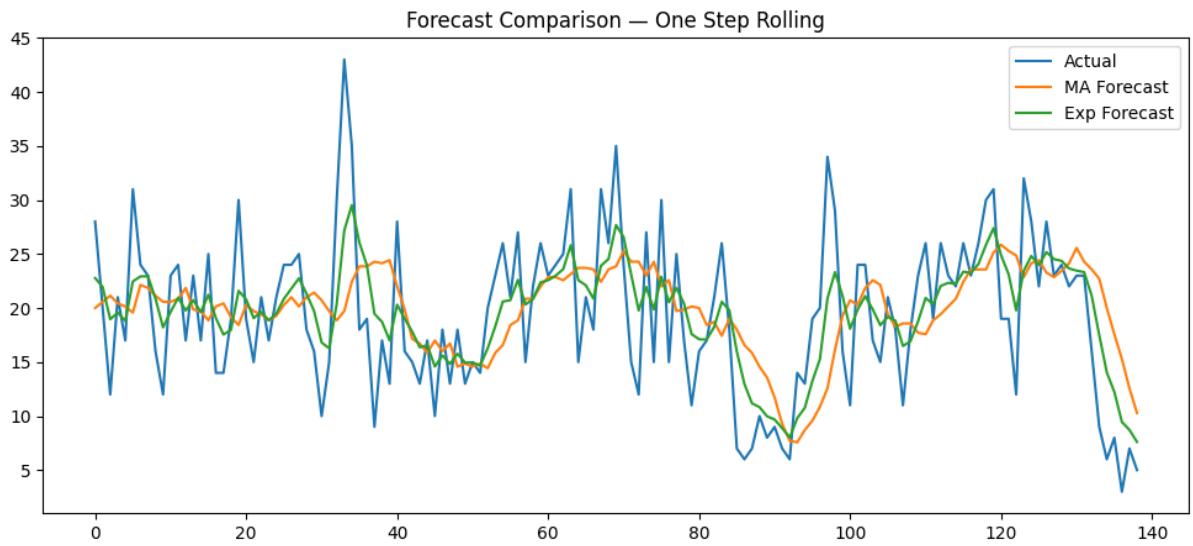
Limitation:

- Does not model trend or seasonality explicitly

Model Accuracy Evaluation

Models were evaluated using standard error metrics.

MODEL	MAE	RSME	MAPE
Moving Average	5.52	6.91	38.26
Exp Smoothing	3.80	4.69	25.12



5.1 Forecast Interpretation

Exponential smoothing showed more stable performance and was selected for operational forecasting. Since it is a level-based model, long-range forecasts stabilize near the latest demand level.

6. Inventory Optimization Modeling

Inventory decisions were optimized using the Economic Order Quantity (EOQ) model to minimize total ordering and holding cost.

6.1 EOQ Model

EOQ balances ordering frequency and holding cost.

EOQ inputs:

- Annual demand
- Ordering cost per order
- Holding cost per unit per year

EOQ output:

- Optimal order quantity
- Reduced total inventory cost

6.2 Safety Stock

Safety stock was computed using:

- Demand standard deviation
- Lead time
- Service level factor (Z value)

Purpose:

- Protect against variability
- Reduce stockout risk

6.3 Reorder Point (ROP)

ROP ensures replenishment is triggered before stockout.

ROP = Lead Time Demand + Safety Stock

This provides continuous replenishment control.